

A Charged Particle on a Flat Magnetic Torus: Clean Return, Flux Quantization, and Signed Landau Multiplets

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Abstract

We fix one normalization for a charged unit-mass particle on a rectangular flat two-torus. At nonzero field, every positive-energy classical orbit has least period $2\pi/|B|$, and every return is maximally clean. Curvature $-iB dx \wedge dy$ admits a Hermitian line bundle exactly when $N = BL_x L_y / (2\pi)$ is integral. On that bundle the physical Bochner Hamiltonian has levels $|B|(n + 1/2)$, each with multiplicity $|N|$ for every flat holonomy. For the fixed positively oriented division vectors, the magnetic translations obey $UV = \zeta_{|N|}^{\text{sgn } N} VU$; hence negative flux carries the conjugate clock-shift representation rather than an erased sign. This round proves the clean return, global quantization, complete Landau ladder, and signed finite-Heisenberg action.

中文摘要

本文统一处理平坦磁二环面的经典回归、全局量子化和朗道能级。经典方程在普适覆盖上可完全积分，守恒回旋中心把所有正能量轨道锁定到共同的最小周期，并说明返回映射在整个能量层上都是恒等，从而得到最大干净而非孤立的周期族。曲率积分给出有符号磁通整数，它既是线丛存在的充要条件，也是线丛次数；任意平坦荷洛诺米只改变连接而不改变能谱。采用带零点半位移的正博赫纳哈密顿量后，升降算符与亏格一的黎曼罗赫定理共同给出完整等距能级及每层磁通绝对值重简并。为避免抹去负磁通，本文固定正向的两条细分格矢，证明磁平移交换相位为单位根的磁通符号次幂，并在每个能级上构造相应的共轭钟移表示。本轮止于完整朗道阶梯和有符号有限海森堡作用。

Keywords: magnetic torus; Bochner Laplacian; Landau levels; flux quantization; magnetic translations; finite Heisenberg group

关键词：磁环面；博赫纳拉普拉斯；朗道能级；磁通量子化；磁平移；有限海森堡群

1 Classical magnetic flow

Let

$$\mathbb{T}^2 = \mathbb{R}^2 / (L_x \mathbb{Z} \oplus L_y \mathbb{Z}), \quad A = L_x L_y,$$

where $L_x, L_y > 0$. On $T^*\mathbb{T}^2$ use lifted coordinates $(q, p) = (x, y, p_x, p_y)$ and the twisted form and Hamiltonian

$$\omega_B = dx \wedge dp_x + dy \wedge dp_y + B dx \wedge dy, \quad H_{\text{cl}} = \frac{1}{2}|p|^2. \quad (1)$$

Fix $B \neq 0$ and write $J(u, v) = (-v, u)$. Our Hamiltonian convention gives

$$\dot{q} = p, \quad \dot{p} = BJp. \quad (2)$$

Theorem 1 (Complete return and clean fixed sets). *On every energy shell $\Sigma_E = \{H_{\text{cl}} = E\}$ with $E > 0$, every orbit has the same least period*

$$T_{\text{cl}} = \frac{2\pi}{|B|}.$$

For $t \notin T_{\text{cl}}\mathbb{Z}$, $\text{Fix}(\Phi_t|_{\Sigma_E})$ is empty. For $t \in T_{\text{cl}}\mathbb{Z}$, the return is the identity and

$$\text{Fix}(\Phi_t|_{\Sigma_E}) = \Sigma_E, \quad \ker(D\Phi_t - I) = T\Sigma_E.$$

Thus every nonzero return is maximally clean. The orbit space of the circle action on $\Sigma_E \cong \mathbb{T}^2 \times S^1$ is a two-torus.

Proof. Let $R_\theta = \exp(\theta J)$. Direct integration of (2) gives

$$p(t) = R_{Bt} p_0, \quad c = q + \frac{Jp}{B} = q_0 + \frac{Jp_0}{B}, \quad q(t) = c - \frac{JR_{Bt} p_0}{B}. \quad (3)$$

This formula is well defined after reducing q modulo the lattice and proves completeness. Since $E > 0$, $p_0 \neq 0$. A phase-space return forces $R_{Bt}p_0 = p_0$, hence $Bt \in 2\pi\mathbb{Z}$; the position then returns as well. This proves both the period and its minimality. At an integral multiple of T_{cl} , formula (3) is the identity for all initial data, so its derivative is the identity. At every other time its rotation has no nonzero fixed momentum. These two observations give the stated fixed sets and clean tangent equality. Finally c modulo the lattice labels an orbit, proving the quotient assertion. $\square$

Geometric meaning. The continuous center c is the classical accidental symmetry. Compact configuration does not isolate trajectories: at a return time the entire three-dimensional energy shell is fixed. This fact is decisive for the Route-A verdict below.

2 Flux integrality and the quantum object

We use units $\hbar = q_{\text{charge}} = m = 1$. A quantum magnetic field means a Hermitian line bundle $\mathcal{L} \rightarrow \mathbb{T}^2$ with unitary connection ∇ whose curvature is

$$F_{\nabla} = -iB dx \wedge dy. \quad (4)$$

Theorem 2 (Sharp flux condition). *A pair $(\mathcal{L}, \nabla)$ satisfying (4) exists if and only if*

$$N = \frac{BA}{2\pi} \in \mathbb{Z}. \quad (5)$$

Its degree is N . For a fixed N , tensoring by a flat line bundle gives the two holonomy parameters and does not change the curvature.

Proof. Chern–Weil theory gives

$$c_1(\mathcal{L})[\mathbb{T}^2] = \frac{i}{2\pi} \int_{\mathbb{T}^2} F_{\nabla} = \frac{BA}{2\pi},$$

which proves necessity. Conversely, for $N \in \mathbb{Z}$ take on the fundamental rectangle the gauge $A_y = Bx$, $A_x = 0$. Identify the vertical sides trivially and identify $x = L_x$ with $x = 0$ by the unitary multiplier $\exp(iBL_x y)$. Its change around the horizontal cycle is $\exp(iBA) = \exp(2\pi iN) = 1$, so the corner cocycle closes and defines the required bundle and connection. Multiplying the two gluing maps by constant phases produces all flat holonomy twists. $\square$

Round-zero certificate: round zero proves complete clean classical return and the sharp global flux condition.

3 Bochner Landau ladder

Put $\Pi_j = -i\nabla_j$. Equations (4) and (5) give

$$[\Pi_x, \Pi_y] = iB, \quad H_B = \frac{1}{2}(\Pi_x^2 + \Pi_y^2) = \frac{1}{2}\nabla^*\nabla. \quad (6)$$

This physical Bochner normalization is important: it includes the zero-point half-shift and is not the unshifted Dolbeault number operator.

Theorem 3 (Spectrum, multiplicity, and holonomy rigidity). *Assume $N \neq 0$. The self-adjoint elliptic operator (6) has*

$$\text{Spec}(H_B) = \{E_n = |B|(n + \frac{1}{2}) : n \in \mathbb{Z}_{\geq 0}\}. \quad (7)$$

Every E_n has multiplicity $|N|$. The level locations and multiplicities are independent of both flat holonomies.

Proof. It is enough to take $B > 0$; negative field is obtained by complex conjugation and interchange of chirality. Define

$$a = \frac{\Pi_x + i\Pi_y}{\sqrt{2B}}, \quad a^* = \frac{\Pi_x - i\Pi_y}{\sqrt{2B}}.$$

Then $[a, a^*] = 1$ and

$$H_B = B(a^*a + \frac{1}{2}), \quad [H_B, a] = -Ba. \quad (8)$$

In the complex structure of the rectangular torus, $\ker a$ is the space of holomorphic sections of the positive degree- N bundle. Riemann–Roch on a genus-one curve and vanishing for the inverse negative-degree bundle give $\dim \ker a = N$, for every flat twist.

The compact elliptic operator has discrete spectrum. Repeated application of a to an eigenvector must terminate, because (8) lowers energy by B while $H_B \geq B/2$. It terminates in $\ker a$, so every eigenvalue is one of (7). Conversely $(a^*)^n$ takes $\ker a$ injectively onto the n th level; the lowering argument makes it surjective. Hence every multiplicity equals N . Complex conjugation replaces this by $|N|$ when $B < 0$. $\square$

4 Finite magnetic translations

Let $M = |N|$, $\sigma = \text{sgn } N$, and $\zeta_M = \exp(2\pi i/M)$. Fix the positively oriented ordered M -division vectors $(L_x/M, 0)$ and $(0, L_y/M)$. Their translations lift projectively to the magnetic bundle. After harmless scalar rephasings, call the corresponding lifts U, V .

Theorem 4 (Finite Heisenberg action). *On every Landau eigenspace one can choose an orthonormal basis $e_0, \dots, e_{M-1}$, with indices modulo M , such that*

$$Ue_j = e_{j+1}, \quad Ve_j = \zeta_M^{-\sigma j} e_j, \quad U^M = V^M = I, \quad UV = \zeta_M^\sigma VU. \quad (9)$$

For $M > 1$ this representation is irreducible; for $M = 1$ it is the trivial one-dimensional boundary case. Flat holonomies change the lifts by phases, not the commutator or the Landau data.

Proof. The bundle cocycle in Theorem 2 makes the commutator of the two fixed-orientation lifted division translations equal to the exponential of the flux through one M -by- M cell. Since

$$\frac{BA}{M^2} = \frac{2\pi N}{M^2} = \frac{2\pi\sigma}{M},$$

this phase is ζ_M^σ ; in particular, negative flux cannot be removed by scalar rephasing. Rephasing does make both M th powers the identity. They commute with H_B , hence act on each eigenspace. Diagonalize V . The commutator cycles its M distinct characters, yielding (9). Since the eigenspace has dimension M , the cycle is a basis and any invariant subspace containing one character contains all. The raising operator intertwines the action, so the same model occurs on every level. $\square$

Round-one certificate: round one adds the complete Bochner Landau ladder, exact multiplicity, holonomy rigidity, and the finite magnetic-translation representation.